

Random Numbers in Use

Cryptography

Mobile communications, e-banking, online payments, ATMs, wireless keys, cybersecurity, and distributed power grid security all depend on true randomness.

Scientific Computing

Monte Carlo calculations, numerical simulations, statistical research, and randomized algorithms require unpredictable numbers for accurate results.

Gaming & Lotteries

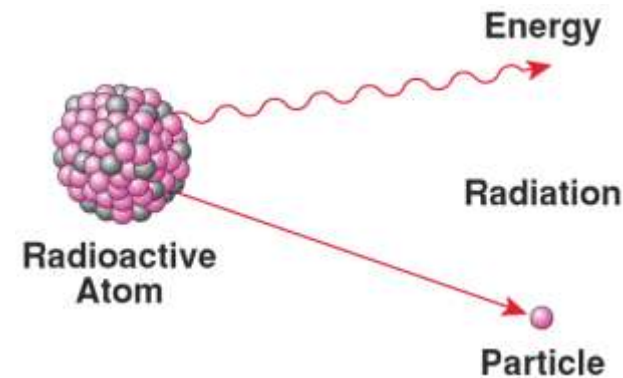
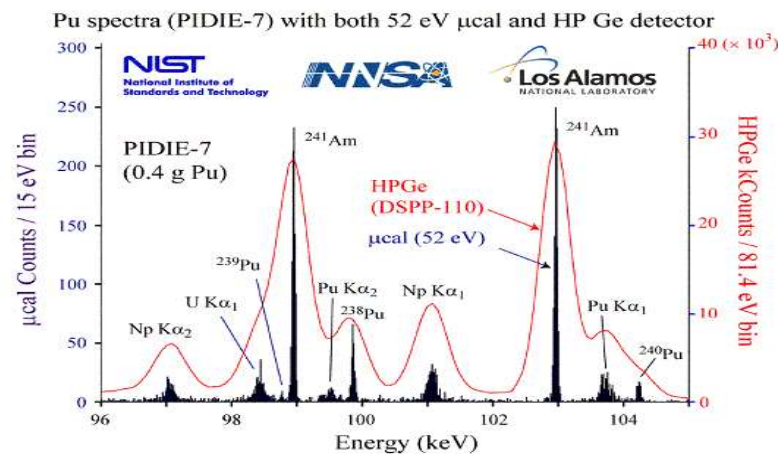
Fair gambling and lottery systems need certified random number generators. Online gambling alone involves an estimated \$6 billion annually in the USA.

Pseudo-random Number Generator

- Pseudo-random number generator:
 - A polynomial-time computable function $f(x)$ that expands a short random string x into a long string $f(x)$ that appears random
- Not truly random in that:
 - Deterministic algorithm
 - Dependent on initial values (seed)
- Objectives
 - Fast
 - Secure

Sources

- The only truly random number sources are those related to physical phenomena such as the rate of radioactive decay of an element or the thermal noise of a semiconductor.
- Randomness is bound to natural phenomena. It is impossible to algorithmically generate truly random numbers.



Good TRNG Design

- Entropy Source:
 - Randomness present in physical processes such as thermal and shot noise in circuits, brownian motion, or nuclear decay.
- Harvesting Mechanism:
 - The mechanism that does not disturb the physical process but collects as much entropy as possible.
- Post-Processing (optional):
 - Applied to mask imperfections in entropy sources or harvesting mechanism or to provide tolerance in the presence of environmental changes and tampering.

Set of Requirements

- The design should be purely digital
- The harvesting mechanism should be simple.
 - The unpredictability of the TRNG should not be based on the complexity of the harvesting mechanism, but only on the unpredictability of the entropy source.
- Compact and efficient design (high throughput per area and energy spent).
- The design should be sufficiently simple to allow rigorous analysis.